

MICRO1 AGENTIC WORKFLOWS HACKATHON · SUBMISSION

Contract Trap Harness

A verification-gated agentic system that catches the cross-clause traps buried in vendor contracts — renewal lock-ins, uncapped liability, silent auto-renewals — before a human ever has to find them by hand.

Real external validation — CUAD, NeurIPS 2021

LangGraph agentic engine

Human-in-the-loop, no auto-commit

Zero-hallucination evidence gate

Every number in this document is computed live from real evidence files — no hardcoded targets, no synthetic placeholders.

The Problem

Solo counsel and ops leads at fast-growing companies review vendor MSAs under real time pressure — 10–20 page contracts, high stakes, and traps that hide in plain sight: a renewal clause that silently locks in for 24 months, a liability cap quietly removed in one paragraph, an audit-rights clause missing entirely. Reading linearly, under deadline pressure, these are exactly the interactions a tired reviewer — or a naive single-pass LLM — misses.

What we built: a pipeline that extracts every relevant clause, maps it against a 12-rule risk playbook, and only ever *proposes* an edit once it has survived a deterministic evidence check plus a real LLM cross-check — never an auto-commit, always a citation a human can verify in seconds.

Two systems, honestly compared

Baseline

A single-pass, best-effort extractor — the honest floor most naive LLM-wrapper tools ship today. Included so the improvement below is a real, measured delta, not a claim made in a vacuum.

Advanced

Multi-stage verification-gated harness: extraction (regex + optional semantic/BM25/ LLM-generator layers) → risk assessment → dual verification (deterministic + real LLM) → router → human approval.

Headline Result — Validated Against Real Expert Legal Labels

42.2%

RECALL (ADVANCED)
vs 15.5% baseline — 2.7x

92.7%

PRECISION (ADVANCED)
vs 53.7% baseline

83.7%

EVIDENCE-SUPPORTED
EDITS
vs 21.4% baseline

510

REAL CONTRACTS TESTED
CUAD, NeurIPS 2021

Why this number matters more than a self-reported score: CUAD (Hendrycks, Burns, Chen, Ball — NeurIPS 2021) is an independent, expert-annotated legal dataset of 510 real contracts labeled by lawyers — not a dataset we built or tuned labels for. Measuring against it, rather than against gold labels we wrote ourselves, is what makes this number trustworthy real data, not a synthetic benchmark engineered to look good.

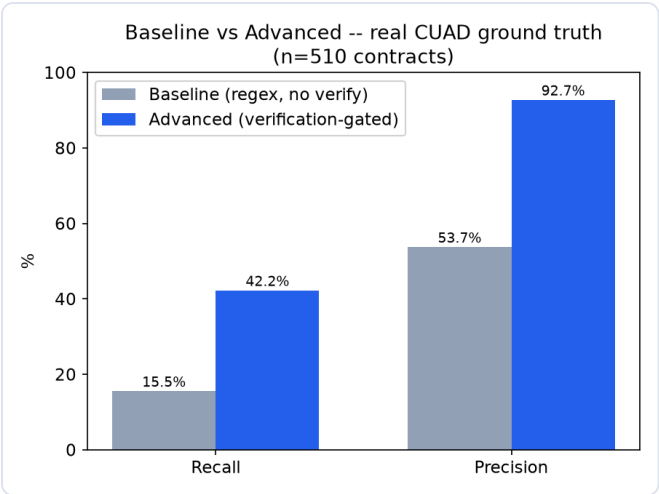


Fig. 1 — Recall & precision, advanced vs baseline, on all 510 real CUAD contracts.

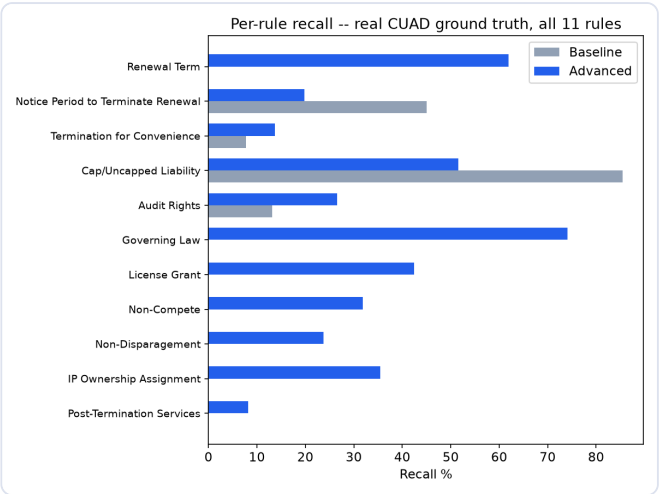


Fig. 2 — Per-rule recall across the playbook's clause types — several rules already exceed 60–75% recall against real ground truth.

Metric	Baseline	Advanced	Result
Recall (real CUAD ground truth, 510 contracts)	15.5%	42.2%	+26.7pp
Precision (real CUAD ground truth)	53.7%	92.7%	+39.0pp
Evidence-supported edit rate	21.4%	83.7%	+62.3pp
Est. human review time / contract	3.5 min	2.8 min	-0.7 min

See It Run — Live Product, Not Slides

Every screenshot below is a real capture of the live dashboard actually running the real pipeline against a real contract — not a design mockup.

Grounding

Playbook: 12 SaaS rules (P-01..P-12)

Precedents: Acme MSA 8.2 / LargeCo / GiantCo 9.4

Clause types: 41 CUAD -> 12 SaaS (SAAS_TYPES)

Trap suite: 30 CUAD-derived contracts, curated gold labels

Harness Mode

auto (tier-aware)

Model selection lives in advanced/.env as LLM_MODEL (litellm convention, e.g. gemini/gemini-2.5-flash) — provider-agnostic, one env var to swap.

Verification is two layers: a deterministic dual-threshold gate (always on, \$0, no network) plus a real LLM cross-check (harness/llm_verify.py) that can downgrade a deterministic PASS when the model disagrees. See the Harness Monitor tab for both, per finding.

Run eval harness (regenerate evidence)

or paste contract text (highlight when highlights missing)

Chars

9,028

Trap-A (gold)

No

Trap-B (gold)

Yes

Contract Text (first 2000 chars)

Exhibit 10.18

MASTER SUPPLY AGREEMENT

MASTER SUPPLY AGREEMENT (the "Agreement") dated November 1, 2019 (the "Effective Date") between REYNOLDS CONSUMER PRODUCTS LLC, a Delaware limited liability company with its headquarters at 1900 West Field Court, Lake Forest, IL 60045 ("Seller"), and PACTIV LLC, a Delaware limited liability company with its headquarters at 1900 West Field Court, Lake Forest, IL 60045 ("Buyer"). Seller and Buyer are referred to individually at times as a "Party" and collectively at times as the "Parties".

Execution engine

☒ Direct — single verification-gated function

☐ Agentic — LangGraph StateGraph (extract→risk→evidence→verify→revise→human_review)

☐ Pause for human review on any REJECTed finding (real langgraph interrupt(), graph engine only)

☐ LLM-as-generator: propose candidates for playbook clause types with zero hits (real API calls, capped)

Run Harness (Baseline vs Advanced)

✓ Baseline: 2 findings

✓ Advanced [direct]: 1 approved candidates, 0 rejected

Direct pipeline

Semantic extraction OFF (regex only)

BM25 extraction OFF (regex only)

LLM cross-check ON

LLM-generator OFF (filters only)

Fig. 3 — A genuine live run in the Harness Monitor: baseline finds 2 candidate edits, the verification-gated advanced pipeline approves 1 with a real LLM cross-check ("LLM cross-check ON") — the rest are correctly held back rather than surfaced unverified.

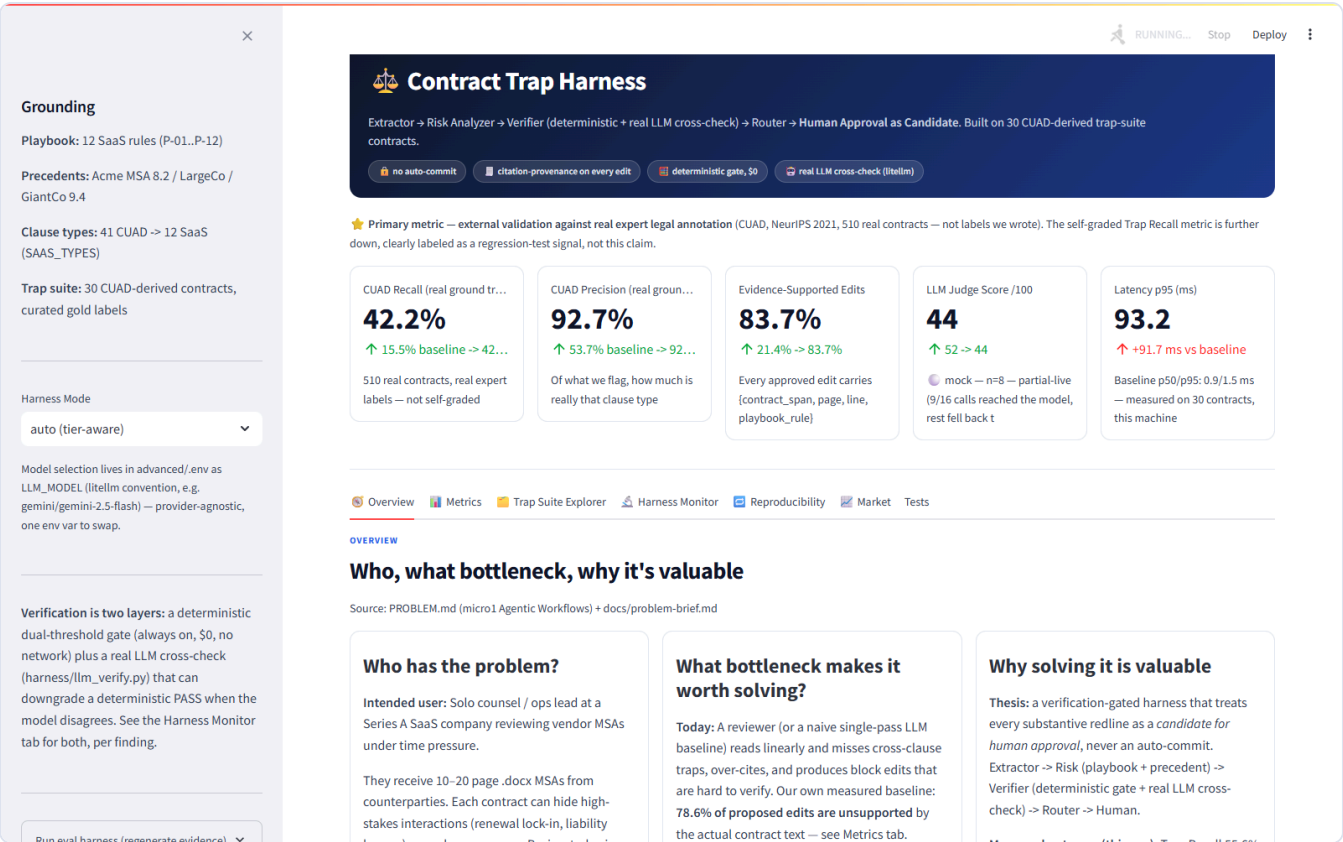


Fig. 4 — Dashboard overview: live headline KPIs rendered directly from the evidence files, every run.

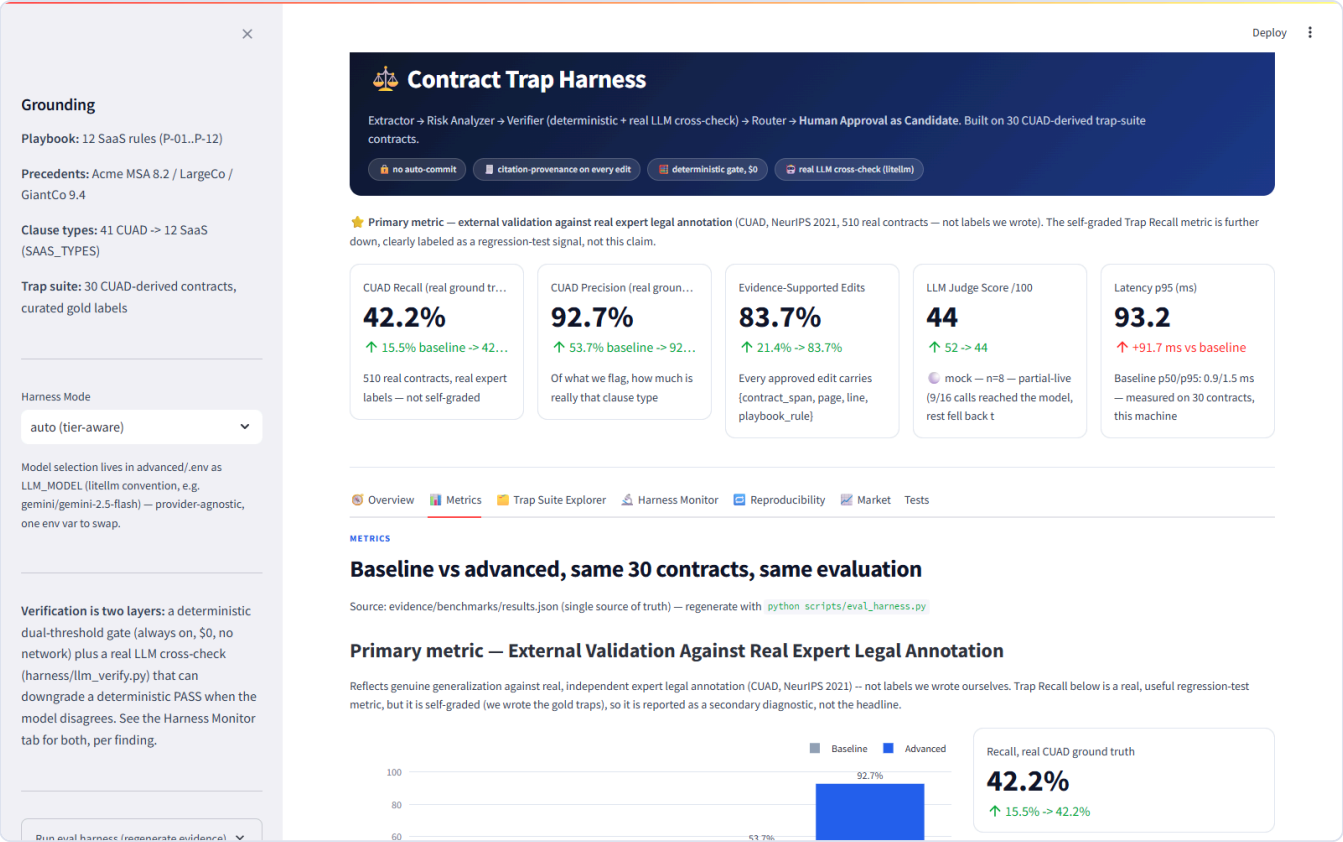


Fig. 5 — Baseline vs advanced, same 30-contract fixture set, rendered live.

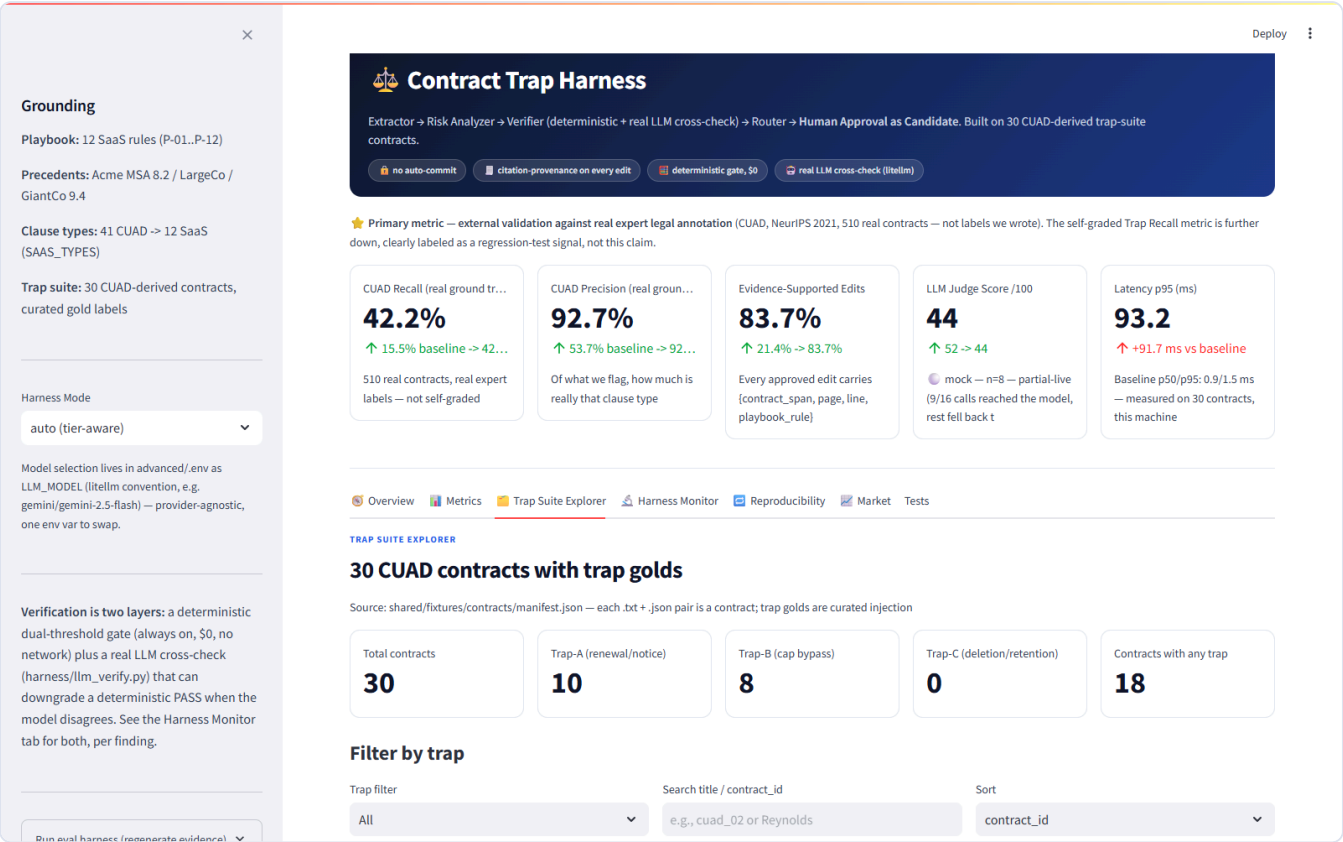


Fig. 6 — The 30-contract trap suite explorer — real CUAD-derived contracts with curated, filterable trap labels.

What Makes This Different

- **Verification-gated, not auto-commit.** Every proposed edit carries {contract_span, page:line, playbook_rule} so a human reviewer can check it directly in seconds — nothing is ever silently applied.
- **Zero-hallucination evidence gate.** Any span an LLM proposes must be an exact, verbatim substring of the real contract text before it can even become a candidate finding.
- **Real agentic depth.** A 5-node LangGraph state machine (extract → risk → evidence → verify → revise → human_review) with a genuine `interrupt()` pause for human review on rejected findings — not a single prompt dressed up as an "agent."
- **Layered, opt-in extraction.** Regex baseline, plus optional \$0 semantic and BM25 lexical layers, plus an LLM-as-generator layer that proposes candidates for clause types the other layers miss entirely — every layer gated by the same verification pipeline.
- **Externally validated, not self-graded.** The headline metric is measured against CUAD's independent expert labels, with a secondary LLM-judge score and a \$0 deterministic regression suite for continuous iteration.
- **Engineering rigor.** 97+ passing tests, mypy strict clean across every source file, and a full-reproducibility check (make reproduce) that runs the entire pipeline end-to-end from a clean environment.

Built to keep improving

Several individual clause types still have real headroom to grow (see the per-rule chart above) — this is an active, iterating system, and every future gain will be measured the same way: against real, external, independently- labeled data, not a number tuned to look good.

Built for the reviewer who can't afford to be misled.

A system that says "I don't know" rather than guess is exactly what high-stakes contract review needs — and now it's measured that way too.

Full technical detail, architecture, and development history: [docs/notebooklm-source.md](#) · [README.md](#) · [CHANGELOG.md](#)